FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VII

JEE (Advanced)-2020

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **One Section**: **Section-A.**

1. ***Section–A (01 – 07, 19 – 25, 37 – 43)*** *contains 21 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

   Section-A (08– 14, 26 – 32, 44 – 50) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer*

   Partial Marks **+1** for each correct option provided no incorrect option is selected.

   Section–A (15 – 18, 33 – 36, 51 – 54) *contains 6 paragraphs. Each paragraph is having 2 multiple choice questions having 4 options with* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

SECTION – A
(One Options Correct Type)

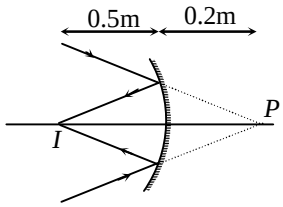
This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A body is projected up so that it can reach maximum height H. What is its height from ground when its velocity is half the maximum velocity?

(A) $\frac{H}{4}$ (B) $\frac{H}{2}$

(C) $\frac{3H}{4}$ (D) $\frac{3H}{8}$

2. A beam of convergent light converges to a point 0.5m in front of the mirror after reflection at convex mirror but in the absence of the mirror the beam converse to a point 0.2 m behind the mirror. The radius of the curvature of the mirror is



(A) 20 cm (B) 50 cm

(C) 66.67 cm (D) 28.5 cm

3. A circular loop of radius *r*, having *N* turns of a wire, in placed in a uniform and constant magnetic field *B*. The normal of the loop makes an angle θ with the magnetic field. Its normal rotates with angular velocity ω in such a way that it maintains the same angle θ with the field, the emf appearing in the loop is

(A) $\frac{\pi NB\omega r^2}{2}$ (B) $\frac{\pi NB\omega r^2}{2}\cos\theta$

(C) $\pi NB\omega r^2$ (D) Zero

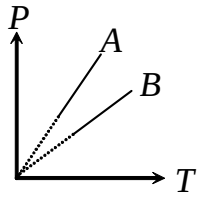
Space for Rough work

4. A point charge θ is located at the center of a hollow spherical conductor having inner radius as R_1 and outer radius R_2. The potential at the inner surface will be

(A) $KQ\left[\frac{1}{R_1}+\frac{1}{R_2}\right]$ (B) $KQ\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$

(C) $KQ\left[\frac{1}{R_2}-\frac{1}{R_1}\right]$ (D) $\frac{KQ}{R_2}$

5. The figure shows the P-T diagram of two different gasses *A* and *B* having same volume but different number of moles n_A and n_B respectively. Then



(A) $n_A > n_B$ (B) $n_A = n_B$

(C) $n_A < n_B$ (D) insufficient data

6. A rigid body of radius R, either hollow or solid, lies on a smooth horizontal surface. The body is pulled by horizontal force acting tangentially from the highest point. The distance travelled by the body in the time in which it makes one full rotation is the same that it will make in one full rotation during pure rolling. The rigid body will be:

(A) Solid sphere (B) Hollow sphere

(C) Circular disc (D) Hollow cylinder

7. A uniform rod of length L (in between the supports) and mass m is placed on two supports A and B. The rod breaks suddenly at length L/10 from the support B. Find the reaction at support A immediately after the rod breaks:

(A) $\frac{3}{20}$mg (B) $\frac{19}{40}$mg

(C) $\frac{\text{mg}}{2}$ (D) $\frac{9}{20}$mg

Space for Rough work

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

8. A solid wooden stick is one fourth submerged in a liquid and is floating. A small mass of negligible valence is attached at the lower end of the stick. Now if the length of stick out of the liquid is one fourth. Then:
   (A) For translational equilibrium of stick mass of attached object should be two times mass of stick.
   (B) For translational equilibrium of stick mass of attached object should be equal to mass of stick.
   (C) Rotational equilibrium of the stick is possible in this situation.
   (D) Rotational equilibrium of the stick is not possible in this situation.

9. Potential energy of a system is given by $U(x)=(x+1)(x+2)$. Then:
   (A) Point $x=-\frac{3}{2}$ corresponds to equilibrium position of the system.
   (B) Point $x=-1$ and $x=-2$ corresponds to equilibrium position of the system
   (C) System is in stable equilibrium position at the $x=-\frac{3}{2}$
   (D) System is in unstable equilibrium position at $x=-\frac{3}{2}$

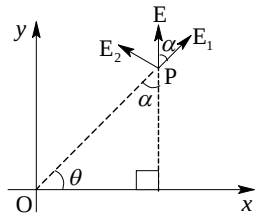
10. Choose the correct options regarding energy densities at different points of a harmonic travelling wave:
   (A) Kinetic energy is maximum at crest
   (B) Kinetic energy is minimum at trough
   (C) Potential energy is maximum at crest
   (D) Potential energy is minimum at trough

11. If the temperature of the bottom of lake of depth of depth $200\,\text{cm}$ is maintained at 4°C and the temperature surrounding remains -2°C for many days. Then:
   (A) The depth of ice formed is 120 cm (B) The depth of ice formed is 80 cm
   (C) The height of water is 80 cm (D) The height of water is 80 cm

Space for Rough work

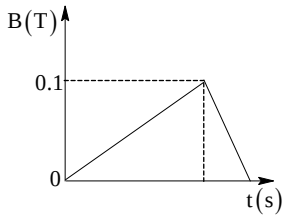
12. Which of the following numbers have three significant digits?
(A) 2.00 (B) 200
(C) 4.01 (D) 0.002

13. A point electric dipole is placed at the origin 'O' and is directed along the positive x-axis. At a point 'P' for away from the dipole, the electric field is parallel to the y-axis. The line 'OP' makes an angle 'θ' with the x-axis, then:



(A) $\tan\theta = \sqrt{3}$ (B) $\tan\theta = \sqrt{2}$
(C) $\tan\theta = \dfrac{1}{2}$ (D) $\tan\theta = \dfrac{1}{\sqrt{2}}$

14. A closed loop of cross-sectional area $10^{-2}m^2$ which has inductance of $10\,\text{mH}$ and negligible resistance is placed in a time varying magnetic field. Figure shows the variation of B with time for the internal 4 seconds. The field is perpendicular to the plane of the loop (Given t = 0, B = 0 and I = 0). The value of current in the loop has:



(A) Maximum value equal to $100\,\text{mA}$
(B) Maximum value equal to $200\,\text{mA}$
(C) Current increases as B increases
(D) Current decreases as B decreases

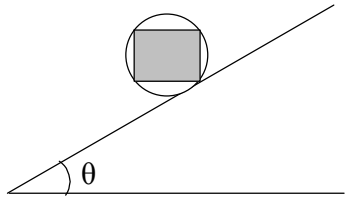
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Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 15 and 16

Four identical rode's each of mass m are welded at ends to form a square and then welded to a ring of same mass m and radius R. If the system is allowed to roll down a rough incline plane of inclination θ, assuming no slipping. Then



15. The acceleration of the system is:

(A) g (B) $\frac{15}{26} g \sin\theta$

(C) $\frac{2}{13} g \sin\theta$ (D) $\frac{2}{26} g \sin\theta$

16. The minimum coefficient of friction for no slipping is:

(A) $\frac{11}{26}\tan\theta$ (B) $\frac{11}{17}\tan\theta$

(C) $\frac{13}{17}\tan\theta$ (D) $\frac{19}{21}\tan\theta$

Space for Rough work

Paragraph for Question Nos. 17 and 18

A small object of mass 2kg is placed on the x-axis in a potential field which potential energy is define as $U = 16\left(x^2 - 2x\right)$ where position on x-axis x is in meter and potential energy U is in joule. If the object is released from origin i.e $x = 0$. Then answer the following questions.

17. Angular frequency of oscillations is:
(A) $2\,\text{rad/sec}$ (B) $4\,\text{rad/sec}$
(C) $8\,\text{rad/sec}$ (D) $12\,\text{rad/sec}$

18. Maximum kinetic energy during oscillations:
(A) $8\,\text{Joule}$ (B) $12\,\text{Joule}$
(C) $16\,\text{Joule}$ (D) $20\,\text{Joule}$

Space for Rough work

Chemistry

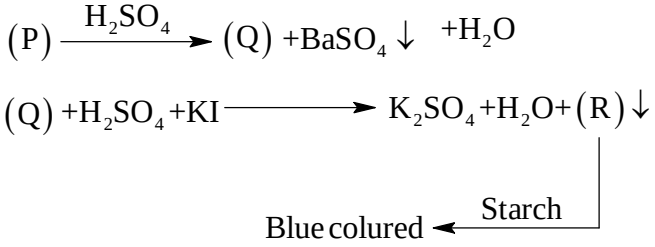
PART – II

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

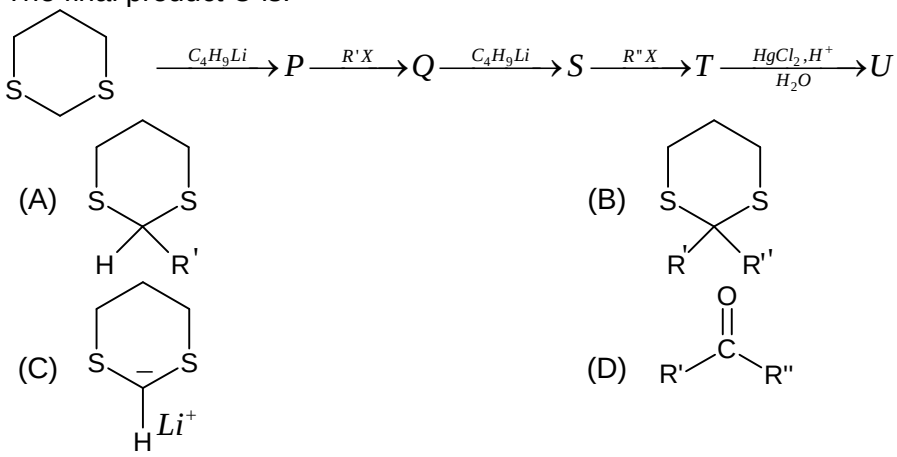
19. A sample of certain mass of an ideal polyatomic gas is expanded against constant pressure of 1 atm adiabatically from volume 2L, pressure 6 atm and temperature 300 K to state where its final volume is 8L. Calculate entropy change (in J/K) in the process. (Neglect vibrational degrees of freedom)
[Take 1L atm = 100J, R=8.0 J/mol/K log 2=0.3, log 3=0.48 log e=2.3]
(A) -3.312 (B) -5.312
(C) -6.312 (D) -7.312

20. If the number of oxygen atom(s) present in hydrated (P) is:



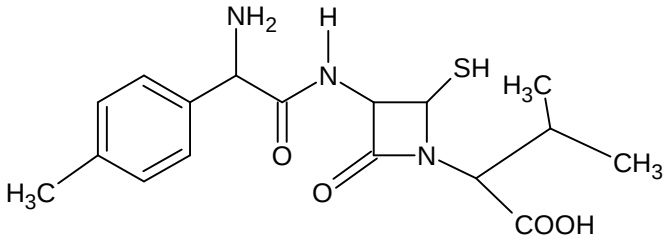
(A) 10 (B) 18
(C) 16 (D) 15

21. The final product U is:



Space for Rough work

22. Find out number of stereogenic center in following compound:

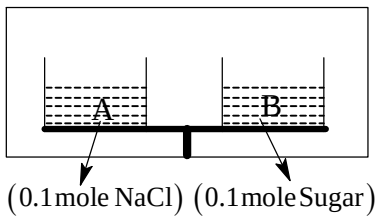


(A) 5 (B) 6
(C) 7 (D) 8

23. When $2.3\times10^{-5}\,g$ steric acid $(M=284)$ was placed on water as a surface film, the film could be compressed to an area of 100 sq cm as a monomolecular layer. Find the cross-section of the acid molecule. (cm^2)

(A) 2×10^{-15} (B) 2×10^{-17}
(C) 2×10^{-19} (D) 2×10^{-21}

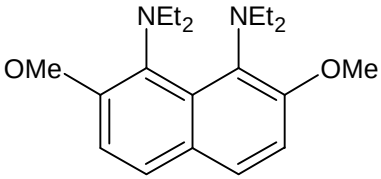
24. Two beakers $\text{A}(0.1\text{mole NaCl in 1kg H}_2\text{O})$ & $\text{B}(0.1\text{mole sugar in 1kg H}_2\text{O})$ is placed in a small sized closed container. The molality of solution of Beaker A changes to $\frac{'x'}{40}$. The value of 'x' will be: (Assume solutions to be dilute)



(A) 17.5×10^{-2} (B) 1.8×10^{-3}
(C) 7.5×10^{-2} (D) 1.8×10^{-4}

Space for Rough work

25. 1,8–Bis(diethyl amino)–2,7–dimethoxy naphthalene is extremely strong base with respect to N, N – Dimethyl aniline.



(A) Protonation relieves the strain operated by lone pair of electrons.
(B) Hydrogen bonding operated in conjugate base
(C) Attachment of +I and +M group with nitrogen and benzene ring.
(D) Both (A) and (B).

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

26. $2\mathrm{CaSO_4}(s) \rightarrow 2CaO(s) + 2SO_2(g) + O_2(g).\Delta H > 0$

Above equilibrium is established by taking sufficient amount of $\mathrm{CaSO_4(s)}$ in a closed container at 1600 K. Then which of the following may be correct option (Assume that $\mathrm{CaSO_4}$ is present in the container in each case)

(A) Moles of $\mathrm{CaO(s)}$ will increase with the increase in temperature.

(B) If the volume of the container is doubled at equilibrium then partial pressure of $\mathrm{SO_2(g)}$ will change at new equilibrium.

(C) If the volume of the container is halved partial pressure of $\mathrm{O_2(g)}$ at new equilibrium will remain same

(D) If two moles of the He gas is added at constant pressure then the moles of $\mathrm{CaO(s)}$ will increase

Space for Rough work

27. Which is / are **correct** statement?

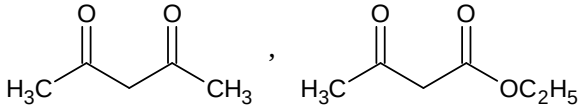
(A) $W_{\text{on the system, adiabatic}} > W_{\text{on the system, isothermally}}$ in an ideal gas compression from same initial state to same final volume

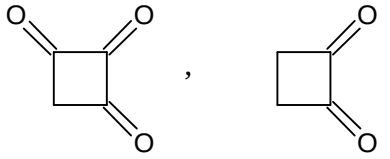
(B) The value of $\gamma \left(\gamma = \frac{C_p}{C_v} \right)$ remains constant for diatomic gas at all temperature

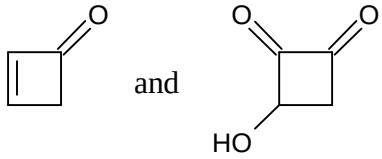
(C) Entropy increases when an ideal gas expanded isothermally.

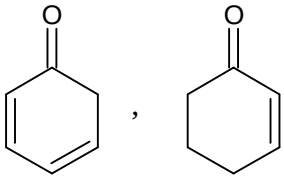
(D) $\Delta_r H$ & $\Delta_r S$ both are + ve for the decomposition of $MgCO_3(s)$.

28. In which of the following compounds, second has lesser enol content than first:

(A) 

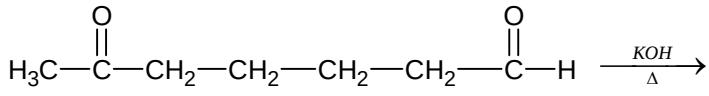
(B) 

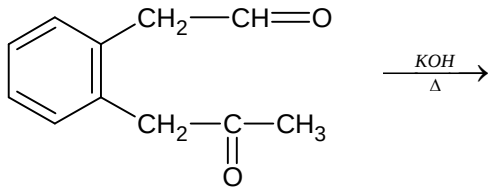
(C) 

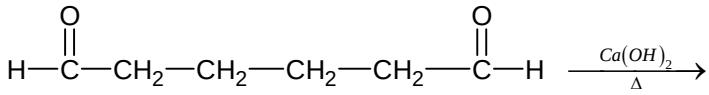
(D) 

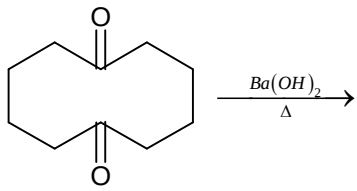
Space for Rough work

29. In which of the following intramolecular aldol condensation reaction more than two products (stereoisomers are not considered) are possible?

(A) 

(B) 

(C) 

(D) 

30. Benzamide and aniline can be distinguished or reacted differently by:
(A) Ferric chloride test (B) Carbyl amine test
(C) Dye azo test (D) $NaOH/Br_2$

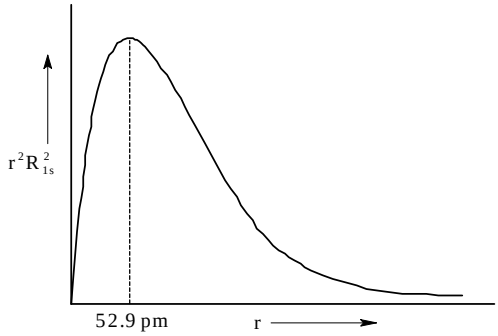
31. Select the correct statement:
(A) Lyophobic colloids are used to protect the lyophilic colloid
(B) Dehydration is used to coagulate the colloidal solution
(C) Rubber is obtained by coagulating the latex
(D) Sometimes heavy rainfall occurs when two oppositely charged clouds meets each other

Space for Rough work

32. Which of the following option is/are correct?

(A) Atomic orbitals are completely described as the regions where the probability of finding the electron is maximum.

(B)



The weighted average of large number of observations for measuring the radius of 1s orbital is greater than $52.9\,\text{pm}$ ($r^2R^2\,dr$ represents the total probability of finding the electron between r and $r+dr$).

(C) The energy of 4s is always lower than 3d for multi electronic atom/ ion.

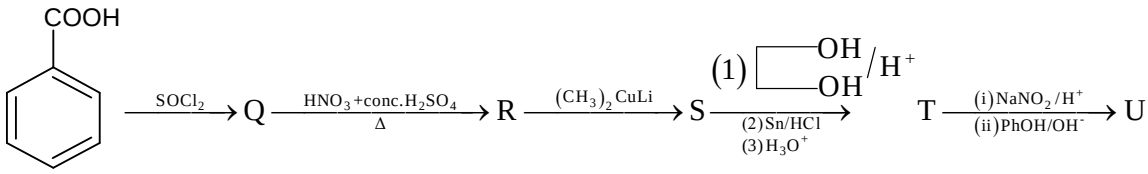
(D) Energy needed to excite an electron from $n=2$ to $n=4$ state is $\frac{25}{28}$ times the energy needed to excite an electron from n=2 to n=5 for a single electron atom / ion.

Space for Rough work

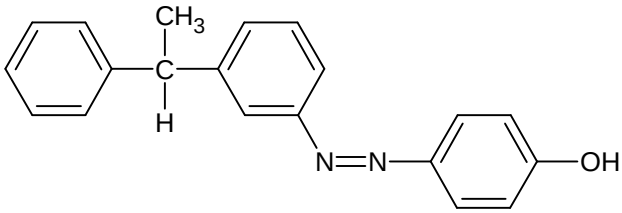
Paragraph Type (One Option Correct)

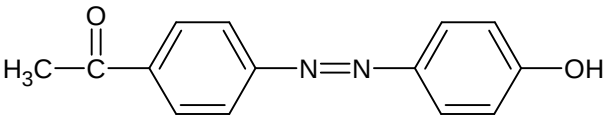
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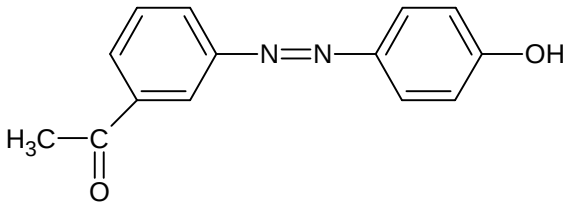
Paragraph for Question Nos. 33 and 34

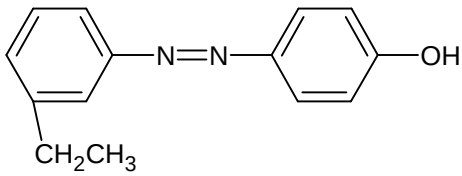


33. The product U is:

(A) 

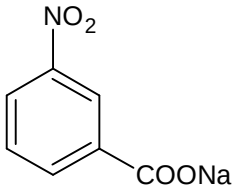
(B) 

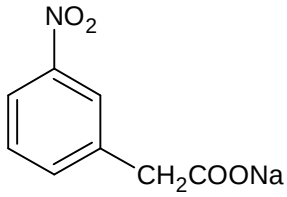
(C) 

(D) 

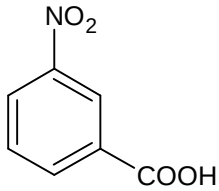
Space for Rough work

34. Which of the following will not be the product/s when "S" on reaction with $Cl_2/NaOH$ or on reaction with $Cl_2/NaOH$ followed by acidification:

(A) 

(B) 

(C) $CHCl_3$

(D) 

Paragraph for Question Nos. 35 and 36

Ligands are broadly classified into two classes classical and non-classical ligands, depending on their donor and acceptor ability. Classical ligands form classical complexes while non-classical ligands form non-classical complexes. Bonding mechanism in non-classical is called synergic bonding:

35. Synergic bonding is absent is:

(A) $\left[Mo(CO)_6\right]$ (B) $\left[Cr(CO)_3(B_3N_3H_6)\right]$

(C) $\left[Sc(CO)_6\right]^{3+}$ (D) $\left[Ni(CN)_4\right]^{4-}$

36. In compound $\left[M(CO)_n\right]^z$, the correct match for highest '$M-C$' bond length for given M, n and z respectively:

	M	**n**	**z**
(A)	Cr	6	0
(B)	V	6	-1
(C)	Ti	6	-2
(D)	Mn	6	$+1$

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **07 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. If a complex number z satisfies $|z|^2 + \frac{4}{|z|^2} - 2\left(\frac{z}{z} + \frac{\bar{z}}{z}\right) - 16 = 0$, then the maximum value of |z| is

(A) $\sqrt{6}+1$ (B) 4

(C) $2+\sqrt{6}$ (D) 6

38. Let S is the focus of parabola $y^2 = 4ax$ and X is the foot of the directrix, PP′ is a double ordinate of the curve and PX meets the curve again at Q. P′Q always passes through

(A) (a, 0) (B) $\left(\frac{a}{4}, 0\right)$

(C) $\left(\frac{2}{a}, 0\right)$ (D) $\left(\frac{a}{2}, \frac{a}{2}\right)$

39. If $g(x) = \frac{2h(x)+|h(x)|}{2h(x)-|h(x)|}$ where $h(x) = \sin x - \sin^n x$, $n \in R^+$, where R^+ is the set of positive real numbers, and $f(x) = \begin{cases} [g(x)], & x \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right) \\ 3, & x = \frac{\pi}{2} \end{cases}$

Where [x] denotes the greatest integer function, then

(A) f(x) is continuous and differentiable at $x = \frac{\pi}{2}$, when $0 < n < 1$

(B) f(x) is continuous and differentiable at $x = \frac{\pi}{2}$, when $n > 1$

(C) f(x) is continuous but not differentiable at $x = \frac{\pi}{2}$, when $0 < n < 1$

(D) f(x) is continuous but not differentiable at $x = \frac{\pi}{2}$, when $n > 1$

Space for Rough work

40. The number of solution of the equation sinx sin2x sin 3x sin4x = $\frac{3}{4}$ is/are

(A) 1 (B) 2
(C) 3 (D) None of these

41. Let a, b $\in$ [0, π], such that $2\cos\left(\frac{\pi}{7}\right)+6\cos a+7\cos b+9\cos\left(\frac{4\pi}{21}\right)=0$ and $2\sin\left(\frac{\pi}{7}\right)-6\sin a+7\sin b-9\sin\left(\frac{4\pi}{21}\right)=0$. Then the value of $\cos(a+b)$ is

(A) $\frac{3}{7}$ (B) $\frac{3}{14}$
(C) $\frac{1}{7}$ (D) None of these

42. If the expression, $2\cos10°+\sin100°+\sin1000°+\sin10000°$ is simplified, then it simplifies to

(A) $\cos10°$ (B) $3\cos10°$
(C) $4\cos10°$ (D) $5\cos10°$

(One or More than one correct type)

This section contains **07** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. Six balls of different colours are to be placed is 3 boxes of different sizes. Each box can hold all the six balls. Number of ways of placing the balls in the boxes so that no box remains empty, is

(A) $3^6-3.2^6+3$ (B) $6^3-6.5^3+{}^6C_2 4^3$
(C) 540 (D) $\left({}^6C_4+\frac{6!}{3!2!}+\frac{6!}{(2!)^3 3!}\right)3!$

Space for Rough work

44. Let $A=\begin{pmatrix} a & b & c \\ b & c & a \\ c & a & b \end{pmatrix}$ If trace $(A)=9$ and a, b, c are positive integers such that $ab+bc+ca=26$. Let A_1 denotes the adjoint of A; A_2 denote adjoint of A_1.....and so on. If value of $det\left(A_4\right)$ is M then

(A) $M=3^{48}$ (B) $M=3^{24}$

(C) Last two digits of M is 61 (D) $Det\left(A\right)=-27$

45. Let $f(x)=\frac{e^{-x}}{1+[x]}$, then

(A) Domain of definition of f(x) is $R-[-1,0)$

(B) Range of definition of f(x) is $\left(-\frac{1}{e},1\right]$

(C) f(x) is one-one in its domain

(D) Graph of f(x) is broken at infinitely many points

46. Set $f(x) = ax^4 + bx^3 + cx^2 + dx + e, a \neq 0,$ is a function such that x = 1, x = 2 and x = 3 are normals to the curve y = f(x) such that f(2) is always greater than f(0), then which of the following are true for f(x)

(i) f(x) has 2 local maxima

(ii) There exist only one value of k such that Rolle's theorem is applicable to f(x) on the interval [0, k]

(iii) f(x) = 0 has two imaginary roots.

(A) Only (iii) and (i) are true (B) (ii) is true and (iii) is false

(C) Only (i) & (ii) are true (D) All are true

Space for Rough work

47. If $I_n = \int_0^{\pi/4} \tan^n x \; dx$, *then*

(A) $I_n + I_{n-2} = \frac{1}{n-1}$

(B) $I_{n+1} + I_{n-1} = \frac{1}{n}$

(C) $\frac{1}{n+1} < 2I_n < \frac{1}{n-1}$, where $n > 1$ is a natural number

(D) $\left(I_{n+2} + I_n\right)(n+1) = 1$

48. $\int \frac{x^2 - 1}{x^3\sqrt{2x^4 - 2x^2 + 1}}\,dx$ is equal to

(A) $\frac{1}{x^2}\sqrt{2x^4 - 2x^2 + 1} + c$

(B) $\frac{1}{x^3}\sqrt{2x^4 - 2x^2 + 1} + c$

(C) $\frac{1}{x}\sqrt{2x^4 - 2x^2 + 1} + c$

(D) $\frac{1}{2x^2}\sqrt{2x^4 - 2x^2 + 1} + c$

49. A fair coin is tossed n times. Let X= then number of times head occurs. If $P(X = 4), P(X = 5)$ and $P(X = 6)$ are in AP, then the value of n can be

(A) 7
(B) 10
(C) 12
(D) 14

50. Die A has 4 red and 2 white faces whereas die B has 2 red and 4 white faces. A coin is flipped once. If it shows a head, the game continues by throwing die A, if it shows tail, then die B is to be used. If the probability that die A is used is $32/33$. When it is given that red throws up every time in first n throws, then find the value of n?

(A) 4
(B) 5
(C) 6
(D) 7

Space for Rough work

Paragraph Type (One Option Correct)

This section contains **02 paragraph** each describing theory, experiment, data etc. **Four questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Question Nos. 51 and 52

If $A_0 = \begin{bmatrix} 2 & -2 & -3 \\ -1 & 3 & 4 \\ 2 & -2 & -3 \end{bmatrix}$ and $B_0 = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$ $B_n = adj\left(B_{n-1}\right)$, $n \in N$ and I is an identity matrix of order 3. C_1, C_2, C_3 represent the column matrix of B_0 as shown, $C_1 = \begin{bmatrix} -4 \\ 1 \\ 4 \end{bmatrix}$, $C_2 = \begin{bmatrix} -3 \\ 0 \\ 4 \end{bmatrix}$ and $C_3 = \begin{bmatrix} -3 \\ 1 \\ 3 \end{bmatrix}$, then answer the following.

51. $\det\left(A_0 + A_0^2 B_0^2 + A_0^3 + A_0^4 B_0^4 + \text{.....upto 10 terms}\right) =$

(A) 1000 (B) –800
(C) 0 (D) –8000

52. $B_1 + B_2 + \text{.....} + B_{49} =$

(A) B_0 (B) $7B_0$
(C) $49B_0$ (D) 491

Space for Rough work

Paragraph for Question Nos. 53 and 54

Saina is to play with Sharapova in a three set match. For a particular set, the probability of Saina winning the set is y and if she wins probability of her winning the next set becomes $\sqrt{y}$ else the probability that she wins the next one becomes y^2. There is no possibility that a set is to be abandoned. R is probability that Saina wins the first set.

53. If R = $\frac{1}{2}$ then the probability that match will end in first two sets is nearly equal to

(A) 0.73 (B) 0.95
(C) 0.51 (D) 0.36

54. If R = $\frac{1}{2}$ and Saina wins the second set then probability that she has won first set as well is nearly equal to

(A) 0.74 (B) 0.46
(C) 0.26 (D) 0.54

Space for Rough work